AYMANE AIT DADS

AI Research Intern | Test-Time Adaptation · Deep Learning · Robust Computer Vision

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams (score 0.86/1.0, $106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by designing staged LoRA fine-tuning pipelines and rigorous ablation studies on a 30B-parameter model, demonstrating the capacity for independent applied ML research. Achieved 1st place in the EURECOM class leaderboard (0.791 AUC) on AI-generated image detection using a 428M-parameter CLIP ViT-L/14 backbone, with a key finding that 1-epoch training outperforms 4-epoch training for out-of-distribution generalization - directly relevant to TTA stability challenges. Combines deep learning, computer vision, and systematic experimental methodology with hands-on experience in model robustness, distribution shift, and ablation-driven research. Aims to contribute to TNO's Collaborative Autonomous Systems department by advancing test-time adaptation methods for perception robustness in autonomous ground, air, and maritime platforms.

CORE COMPETENCIES

Test-Time Adaptation | Deep Learning | Model Robustness | Distribution Shift | Ablation Studies | Computer Vision | Autonomous Systems | Out-of-Distribution Generalization

WORK EXPERIENCE

Orange Maroc
Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built and evaluated **Random Forest and K-Means models** on 100,000+ fiber-optic network records to classify network quality across thousands of nodes, applying systematic model evaluation methodology.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with outputs adopted directly by the network optimization team for infrastructure decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering, demonstrating applied ML deployment in an operational environment with real performance constraints.
- Delivered stakeholder presentations translating model outputs into **actionable maintenance recommendations**, bridging technical findings and real-world operational decision-making.

PROJECTS

AI-Generated Image Detection - EURECOM ImSecu + NTIRE 2026 @ CVPR
EURECOM Course + International Benchmark · 2025 · Ranked 1st in Class · CVPR Submission

- Ranked **1st in EURECOM class** (private Kaggle leaderboard, 0.791 AUC) fine-tuning CLIP ViT-L/14 (428M params) with dual learning rates and unfreezing strategies across last 4/6/8 transformer blocks; also submitted to **NTIRE 2026 @ CVPR** CodaBench international benchmark.
- Discovered that **1-epoch training (0.793 AUC) outperforms 4-epoch training (0.767 AUC)** for out-of-distribution generalization - a stability and adaptation insight directly analogous to test-time adaptation challenges; applied 10-view TTA ensemble weighted by validation AUC over 250K training images.

PyTorch · OpenCLIP · ViT · FP16 · AdamW

NVIDIA Nemotron Model Reasoning Challenge - Kaggle (In Progress)
Kaggle Competition · In Progress (June 2026) · Ranked 17th / 3,000+ Teams · $106K+ Prize Pool

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, $106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context).
- Performed **task-level error analysis** across 5 reasoning categories (unit conversion, cipher, symbolic, gravity, bit manipulation); built a diagnostic evaluation pipeline revealing that reasoning over-expansion - not incorrect reasoning - was the primary failure mode, demonstrating the value of targeted diagnostic work over brute-force data scaling.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM · Triton

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM
Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Image Security, NLP, Statistics, Cloud Computing

CPGE - Mathematics & Physics - Ibn Timiya
2021 - 2023

SKILLS

Deep Learning & Model Adaptation: PyTorch | LoRA / PEFT | TRL / SFTTrainer | OpenCLIP / ViT | Ablation Studies | FP16 / BF16 Mixed Precision | Hugging Face Transformers

Computer Vision & ML Research: CLIP | CNN / DenseNet | Scikit-learn | TTA (Test-Time Augmentation) | Ensemble Methods | Anomaly Detection | Kaggle GPU / CUDA

Data Science & Tools: Python | Pandas / NumPy | Feature Engineering | Power BI | Git / Linux | Docker (basic) | SQL